From Concern Detection to Promotion Control: A Controlled Study of Fail-Closed Research Governance

Anonymous ACL submission

Abstract

Research workflows can produce complete-looking manuscripts and run bundles while leaving scientific defects unresolved. We study *promotion governance*: whether an artifact bundle may advance and which workflow stage must repair it. We introduce a mechanically certified benchmark with 120 base bundles, 600 registered fault cases, and 120 clean controls. A fail-closed artifact audit made zero false promotions and retained all clean controls; presence-only and advisory-only policies promoted every fault case. Under perfect audit-derived routing, owner-scoped repair made all 600 fault cases acceptable to the same gate without changing clean controls, while 360 repairs exactly reconstructed the paired clean artifact tree. Three self-recorded local-model runs over short Results fragments abstained on every case and are reported separately as a protocol diagnostic. The findings establish conformance to one generated suite and its executable fault contracts, not population-level effectiveness or naturalistic reviewer generalization.

1 Introduction

Agent Laboratory connects literature review, experimentation, and reporting (Schmidgall et al., 2025), and PaperBench evaluates executed research replication (Starace et al., 2025). Completion, however, does not imply scientific readiness. Unsound generated papers can receive favorable automatic reviews (Jiang et al., 2026), and Dycke and Gurevych (2026) find that injected reasoning flaws need not materially change tested reviewer outputs.

This gap is also a control problem. A system may detect missing evidence yet still accept a bundle; a checklist may find required files while missing contradictions; a conservative model may avoid false acceptance by rejecting everything. We call the policy connecting concerns to allowed workflow transitions *promotion governance*. It should block unsupported promotion, retain clean bundles, expose decision evidence, and route repair to the stage that owns the defect.

We ask: **within a fixed executable artifact contract, how do artifact visibility and fail-closed concern-to-decision binding affect false promotion, clean retention, and repair routing?** We contribute (i) an artifact-bundle task covering decisions, concerns, evidence paths, and repair ownership; (ii) a mechanically certified benchmark with family-separated development and test partitions, grouped base variants, exact-hash duplicate screening, and independent mutation replay; and (iii) a controlled component study with node-owned recovery over 720 test cases. Claims are limited to registered artifact faults in the generated suite.

2 Related Work

Agent Laboratory composes agents across the research workflow (Schmidgall et al., 2025), whereas PaperBench evaluates replication from paper understanding through executed artifacts (Starace et al., 2025). BadScientist and counterfactual reviewer evaluations expose failures in automatic scientific judgment (Jiang et al., 2026; Dycke and Gurevych, 2026). CLAIM-BENCH evaluates claim extraction, evidence extraction, and their links (Javaji et al., 2025). ARA and REPRO-Bench assess reproducibility from workflow or reproduction artifacts (Riehl et al., 2026; Hu et al., 2025), while SAGE localizes experiment failures to intervention levels (Ma et al., 2026). The present task joins artifact consistency, decision binding, and repair localization; a passing gate means contract conformance, not scientific truth.

3 Task and Benchmark

Each case is a portable research-run tree. A system returns one of `promote`, `needs_review`, `downgrade`, or `block`; typed concerns with severities and artifact paths; and repair owners.

Gold and mutation metadata remain scorer-side. Exact case coverage, predictions, the suite, a contemporaneous TypeScript source snapshot, and the dependency lock are SHA-256 bound. A fail-closed policy forbids `promote` when an unresolved blocker exists. We also measure *concern–acceptance conflict*: promotion among faulty predictions that actually emit a blocking concern.

The benchmark comes from a versioned mutation registry. Every entry specifies an artifact, precondition, deterministic transformation, expected decision, blocker, and repair owner. A separately implemented oracle reconstructs each case from its clean control, replays the mutation, checks before/after hashes, and derives gold. Registry, split, suite, gold, and oracle-report hashes are bound before scoring. This is executable contract gold, not independent human judgment of scientific correctness.

Development and test use disjoint fault families; all variants of a base stay in one partition. Exact source hashes screen byte-identical duplicates but do not establish substantive source independence. The 120-case development set contains 24 bases and four fault families. The 720-case test set contains 120 bases, each with one clean control and five faults: claim–evidence conflict, declared-versus-executed budget mismatch, missing repeated-run provenance, figure-audit summary-state contradiction, and claim-contract flag contradiction. The latter two are metadata-state faults and account for 240 of 600 faults. Both partitions reuse semantically equivalent generator templates, and no pre-outcome freeze prevents tuning to the generator; this is not a held-out-source generalization design.

Systems. **Always promote** preserves readiness. A **presence checklist** checks required files but not preserved-file field mutations. An **advisory audit** copies the full audit's concerns and owners but forcibly retains `promote`. The **fail-closed audit** applies artifact checks and blocks or downgrades unresolved findings. These deterministic conditions isolate contract mechanisms rather than estimate effectiveness against an independent governor.

Qwen3.5-35B-A3B is the model identity cited for the fragment-model diagnostic (Qwen Team, 2026). The local runner separately self-records that identifier as its requested and resolved model for short Results fragments. Every six-case sibling group receives the same 96–98-character fragment under opaque request identifiers. Prompts omit case-specific paths, mutation labels, gold, and scored concern and owner vocabularies. Three complete local-runtime trials contain 2,160 predictions and self-record 324,216 input and 131,976 output tokens. This protocol cannot expose the artifact faults, so it is an execution and abstention diagnostic, not a learned-governor comparator.

4 Evaluation

The primary metric is false promotion over faulty cases. Secondary metrics are four-way decision accuracy, macro-F1, clean promotion, blocker F1, exact owner match, and artifact trace coverage. For registered benchmark concerns, a trace counts only when every concern-specific artifact target is referenced and each path resolves inside the case tree. Unregistered model concern codes retain an existence-only trace diagnostic and are interpreted separately. We report finite-suite rates. The scorer also makes 5,000 deterministic `base_bundle_id` resamples and reports empirical 2.5th/97.5th percentiles as identifier-sensitivity intervals. Its seed is the first 32 bits of SHA-256 over the null-delimited analysis identity. These are not confidence intervals for unseen bundles and do not determine the fixed-suite gate.

Four post-hoc criteria, recorded as H1–H4 for machine compatibility, are used: H1 requires the checklist-minus-audit false-promotion difference to be at least .20; H2 requires audit concern–acceptance conflict at most .05; H3 requires clean promotion at least .90; and H4 requires the audit-minus-always-promote owner-match difference at least .15. The machine-readable gate remains false for prospective evidence because the design is post hoc, even when its conformance blockers are empty.

Recovery starts from fail-closed predictions. Predicted decisions select cases and predicted owners select experiment, analysis, or figure-audit adapters. Adapters receive neither registry labels nor clean siblings. Outside the adapters, mutation family asserts owner–family consistency and repaired-case gold is set to `promote` before the same audit reruns. Gold then scores routing and restoration. A manifest binds changed paths, adapter revisions, before/after tree hashes, and source prediction hashes. This evaluates repair under observed perfect routing, not recovery from an erroneous decision or owner.

5 Results

The deterministic comparison isolates concern-to-transition binding. Always promote, the checklist, and the advisory audit accepted all 600 faults. The advisory policy nevertheless achieved perfect blocker F1 and owner match; it emitted blockers on 480 faults and promoted all 480, giving conflict rate 1.0. The fail-closed audit made zero false promotions and retained all 120 clean controls. Its fixed-suite false-promotion reduction against the checklist was 1.0 and its decision-accuracy increase was .833. These values follow the registered mechanisms, not an independently designed comparator.

Table 1 summarizes these deterministic rows. H1–H4 observed values were 1.0, 0.0, 1.0, and 1.0, with sensitivity intervals $[1, 1]$, $[0, 0]$, $[1, 1]$, and $[1, 1]$. Their eligible counts were 600, 480, 120, and 600; each report also records 600 faulty and 720 total suite observations. All criteria passed as post-hoc finite-suite checks. The prospective evidence gate remained false because the design was post hoc.

The fragment model blocked every case in every trial. Its pooled decision accuracy was .500, macro-F1 .167, false promotion .000, clean promotion .000, blocker F1 .000, owner match .000, and trace coverage 1.000; that trace metric checks only whether the requested constant path exists. Blanket abstention under an artifact-blind fragment prompt does not measure evidence grounding, clean utility, or learned-governor effectiveness.

Recovery covered all 600 faults and reaccepted every repaired case; no clean control changed. Table 2 shows that 360 repairs exactly matched clean trees. Claim–evidence repair restored the unique evidence link without changing its citation metadata, and repeated-run repair used new trial identifiers, so both remained byte-distinct. The deterministic adapter and repaired artifacts make these operations inspectable.

Three earlier schema-1.0 attempt ledgers self-report stale-running recovery counts of 499, 479, and 479 responses before completing 720 requests each. They record prior and observed hashes but do not preserve the corresponding preimage files, independently verify their recorded prefix contents, or encode an external cause. They therefore support only a bounded claim about runner-recorded resume behavior, not power-loss attribution, provider independence, or statistical replication.

6 Conclusion

On one executable controlled benchmark, fail-closed concern binding prevented all registered false promotions while retaining every clean control; advisory concerns alone did not constrain transitions. Under perfect audit-derived routing, owner-scoped repair restored same-gate acceptance for every fault and exactly reconstructed 60% of clean trees. These are contract-conformance results, not broad claims about scientific validity or reviewer superiority.

7 Limitations

The benchmark uses generated faults, one artifact schema, one generator, and contract-derived gold without independent scientific annotation. Development and test are fault-family separated but not substantively source-independent. The deterministic audit was designed for the same contract, and no competitive artifact-aware learned reviewer receives matched inputs. The model diagnostic uses one local model and an artifact-blind fragment prompt; self-recorded receipts do not establish external identity or independent sampling. The frozen suite does not bind the historical corpus-generator source at execution time; oracle replay establishes artifact consistency, not that historical identity. Provider resume evidence uses earlier ledgers without retained preimages. Recovery starts from perfect routing, is scoped to registered adapters, and achieves only 60% exact reconstruction. Same-gate reacceptance cannot establish semantic repair, naturalistic validity, or robustness to unmodeled faults. Full-text citation approval remains a separate human-authority requirement.

8 Ethical Considerations

A gate cannot establish scientific truth; it can only enforce consistency among inspected evidence, concerns, and transitions. False promotion can amplify unsupported claims, while excessive blocking can suppress valid null or negative results. Deployments should report clean retention and unresolved items, preserve appeal and human review, and exclude credentials, personal data, private paths, and non-redistributable sources from public bundles.

System	Trials	Dec. acc.	Macro-F1	False prom.	Clean prom.	Blocker F1	Owner	Trace
Always promote	1	.167	.071	1.000	1.000	–	.000	–
Presence checklist	1	.167	.071	1.000	1.000	–	.000	–
Advisory audit	1	.167	.071	1.000	1.000	1.000	1.000	1.000
Fragment model	3	.500	.167	.000	.000	.000	.000	1.000
Fail-closed audit	1	1.000	1.000	.000	1.000	1.000	1.000	1.000

Table 1: Controlled-suite results. Each deterministic row has 600 faulty and 120 clean observations. The separate artifact-blind fragment diagnostic pools 1,800 faulty and 360 clean prediction-observations from three receipt-distinct executions that are not statistical replicates. A dash denotes an undefined metric when no concern is emitted.

Fault family	Reaccepted	Exact tree
Claim–evidence conflict	120/120	0/120
Executed-budget mismatch	120/120	120/120
Repeated-run provenance	120/120	0/120
Figure-audit summary state	120/120	120/120
Claim-contract flag state	120/120	120/120

Table 2: Same-gate reacceptance and exact restoration under perfect audit-derived routing.

References

Nils Dycke and Iryna Gurevych. 2026. Automatic reviewers fail to detect faulty reasoning in research papers: A new counterfactual evaluation framework. *Transactions of the Association for Computational Linguistics*, 14:465–488.

Chuxuan Hu, Liyun Zhang, Yeji Lim, Aum Wadhwani, Austin Peters, and Daniel Kang. 2025. REPRO-Bench: Can agentic AI systems assess the reproducibility of social science research? In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23616–23626, Vienna, Austria. Association for Computational Linguistics.

Shashidhar Reddy Javaji, Yupeng Cao, Haohang Li, Yangyang Yu, Nikhil Muralidhar, and Zining Zhu. 2025. Can AI validate science? benchmarking LLMs on claim →Evidence reasoning in AI papers. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 2355–2379, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Fengqing Jiang, Yichen Feng, Yuetai Li, Luyao Niu, Basel Alomair, and Radha Poovendran. 2026. BadScientist: Can a research agent write convincing but unsound papers that fool LLM reviewers? In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24712–24727, San Diego, California, United States. Association for Computational Linguistics.

Jie Ma, Binfei Chu, Jie Gao, Jinlu Zhang, Yiwei Ma, Yi Tan, Jiayi Ji, Xiaoshuai Sun, and Rongrong Ji. 2026. One reflection is not enough: Self-correcting autonomous research via multi-hypothesis failure attribution. *Preprint*, arXiv:2606.31478. ArXiv version v1, accessed 2026-07-16.

Qwen Team. 2026. Qwen3.5: Towards native multimodal agents.

Kevin Riehl, Andres L. Marin, Nikofors Zacharof, Fan Wu, Patrick Langer, Robert Jakob, Anastasios Kouvelas, Georgios Fontaras, and Michail A. Makridis. 2026. Ara: Agentic reproducibility assessment for scalable support of scientific peer-review. *Preprint*, arXiv:2605.02651. ArXiv version v2, accessed 2026-07-16.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. 2025. Agent laboratory: Using LLM agents as research assistants. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 5977–6043, Suzhou, China. Association for Computational Linguistics.

Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, Rachel Dias, Evan Mays, Benjamin Kinsella, Wyatt Thompson, Johannes Heidecke, Amelia Glaese, and Tejal Patwardhan. 2025. PaperBench: Evaluating AI's ability to replicate AI research. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 56843–56873. PMLR.